



Quality Assessment of Surface Current and Wave Measurements from the High-Frequency Radar System in the Gulf of Trieste

Section 3:

Development of new quality-control (QC) algorithms for the detection of anomalies that remain undetected by existing algorithms

1 Introduction

Based on the findings of the Work Project 2: Evaluation of the effectiveness of existing quality control (QC) algorithms for measurement data, hereafter *wp2*, the objective of this work package is to develop or adapt new procedures for the detection and flagging of anomalies and artefacts in the HFR GOT for the 2021–2026 time series, that are not addressed by the current algorithms.

In particular, the new QC algorithms should be capable of detecting non-physical divergences and convergences in the flow field. The algorithms should enable reproducible processing and the documented removal or flagging of suspicious values. It should also be investigated whether unsupervised artificial intelligence algorithms can be used to distinguish clusters of non-physical artefacts from other situations.

Note 1: All scripts and corresponding results are provided alongside this document.

2 Selection of the types of anomalies/artefacts to be addressed using new algorithms, based on the report from WP2

The HFR-NAdr network publishes data through the EU HFR Node, which applies a standardized QC suite to all member radar systems before publication to ERDDAP. QC is applied at two product levels: *Radial product*: (per-station: AURI, IZOL, PIRA, TRI1), and *Total product*. A vector can therefore be flagged at the radial stage, the total-combination stage, or both.

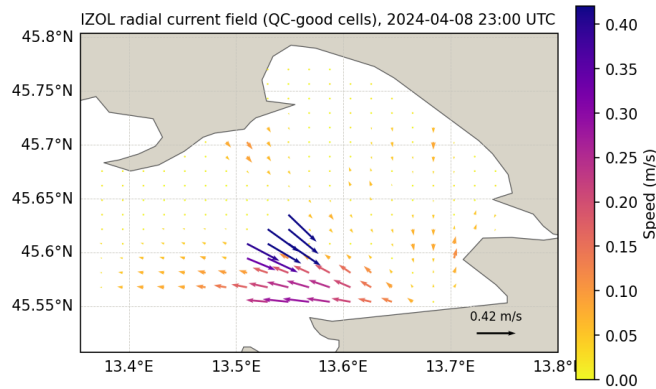


Figure 2: Velocity field example from HFR-NAdr-IZOL at 2024-04-08 23:00UTC, where all QC algorithms passed with QCflag=1.

In wp2, tasks 2 and 3 established two different failure modes for the existing QC:

- Where an existing test is well-matched to a real problem, RDCT_QC against station outages and low radial count, GDOP_QC against the AURI/PIRA baseline geometry, CSPD_QC/VART_QC/MDFL_QC against extreme or unstable individual vectors.
- Where the failure has no matching existing test, it passes as “good data” regardless of how implausible it looks. Figure 2 is a concrete example.

Overall, the existing EU HFR Node QC does its intended job well within its own scope, i.e. geometry and count-based rejection. However, it was not designed to detect physical-consistency issues, including non-physical divergence, direction reversals, and inconsistencies between neighboring vectors, which are the focus of this work.

The new algorithms should detect unphysical divergences/convergences (two small opposing patches side by side, or a patch flowing against its surroundings), unphysical shear (a narrow strip flowing to both sides, direction flipping between adjacent cells), single-cell reversals/spikes (one cell opposite to or much stronger than all its neighbours), and ideally Bora or Scirocco outbursts (strong coherent jets with a hard edge). Strong outbursts are physically real, so they are not treated as errors.

3 Development or adaptation of QC algorithms for the selected types of anomalies. Documentation of the methods and parameters used

Note 2: No new data were downloaded; this analysis was performed entirely on the already unified and published archive. All five products share one regular 20×22 lat/lon grid (1.5 km) and the same half-hour time steps.

For this analysis, two algorithms were developed and tested to detect the selected types of anomalies: k-means clustering, and convolution with different types of kernels and a sliding window.

In both analyses, a cell’s value was considered valid only if its QC flag was good (flag 1 or 2 in Table 3, wp2) of the existing EU HFR Node QC. Cells with bad QC flags (3 or 4) were treated identically to true no-coverage NaN values and were therefore not used as input. The new flags look for errors the existing QC lets through, so they add to the existing QC and do not replace it.

3.1 K-means clustering

For this analysis, the script *cluster_regimes.py* was developed, while the outputs are *cluster_summary_<product>.csv* and *cluster_<product>_k<k>_cluster<c>.png*.

The script ran k-means, with k up to 8, over the full 2021-present HFR archive: once for the Total product (one feature vector per timestep, built jointly from EWCT(u)+NSCT(v)), and once for each radial station (AURI, PIRA, TRI1, IZOL; one feature vector per timestep per station, built from RDVA only). It then rendered the mean field of each cluster, so that it could be assessed whether the clusters corresponded to recognizable circulation regimes (Bora, Scirocco, Tramontana) and/or whether a divergence-dominated cluster emerged.

The results were not satisfactory with respect to the intended objective. For example, for the Total product, for $k = 2$, the clustering primarily separated the observations according to flow direction, resulting in two clusters corresponding to a northeastward (NE) and a southwestward (SW) flow, respectively (see example in Figure 3 (left)). However, increasing the number of clusters did not lead to the identification of additional clusters representing distinct flow patterns or behaviours. Instead, the algorithm continued to subdivide flows exhibiting the same direction according to their magnitude. Consequently, the initial separation into NE and SW-directed flows progressively evolved into clusters representing strong and weak SW flow, as well as strong and weak NE flow (see example for $k = 6$ in Figure 3 (right)). This behaviour was consistently observed also for the Radial products.

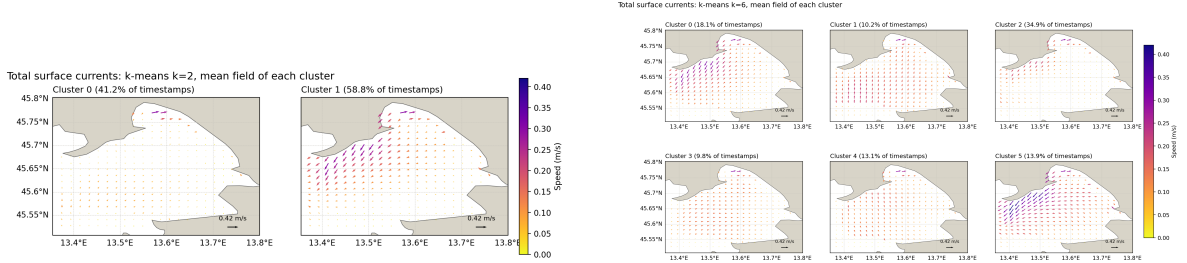


Figure 3: k-means clustering with $k = 2$ (left) and $k = 6$ (right) on Total surface current field.

The key result was that the divergence events disappear in cluster-mean fields, because they occur at a different place each time. The main limitation can be attributed to the averaging: each cluster is represented by the average field of all the time points within it, whereas divergences occur at different locations and times, so they cancel each other out in the average.

Given that this analysis did not render the expected results, it was not pursued further, and an alternative approach was explored.

3.2 Sliding-window kernels + Isolation Forest

For this analysis, the scripts *kernel_features.py*, and *kernel_clustering.py* were developed, alongside the outputs `kernel_anomaly_map_<product>[_3x3].png`, `kernel_anomaly_monthly_<product>[_3x3].png`, `kernel_frequency_<product>[_3x3].png`, and `kernel_featuremaps_<product>_<time>.png`.

Unphysical divergence, convergence and shear appear as small, local features, a few neighbouring cells whose velocities point in opposite directions or change abruptly over one grid cell. Because these features occur at a different place each time, they average out when the whole current field is analysed at once. Therefore, the field was analysed locally, using a sliding window: a small square of 3×3 or 4×4 grid cells that is moved across the whole grid one cell at a time, at every half-hourly time step. Each window position is treated as a separate sample.

A window is only used when all of its cells are marked as good by the existing QC, so the method works only on data that the current QC already accepts.

Within each window four *half-split kernels* were computed (see Table 1 and Figure 4). A kernel is a fixed pattern of weights that divides the window into two halves, A and B, and returns the mean velocity of half A minus the mean velocity of half B.

The four kernels compare north with south (TB), east with west (LR), and the two diagonal halves (DR and UR).

Kernel	Half A	Half B	4 x 4 cells	3 x 3 cells
TB (top/bottom)	north rows	south rows	8 / 8	3 / 3 (middle row weight 0)
LR (left/right)	east columns	west columns	8 / 8	3 / 3 (middle column weight 0)
DR (diagonal ↘)	NE triangle	SW triangle	6 / 6 (diagonal 0)	3 / 3 (diagonal 0)
UR (diagonal ↗)	NW triangle	SE triangle	6 / 6 (diagonal 0)	3 / 3 (diagonal 0)

Table 1: Definition of the kernel configurations and corresponding cell groups.

For the total product the kernels were applied separately to the eastward EWCT (u) and northward NSCT (v) components, which gave eight values per window. For the radial products they were applied to the radial velocity, which gave four values.

The kernel values have a direct physical meaning. A large positive LR value on u or TB value on v means that the two halves move apart (divergence), and a large negative value means that they move towards each other (convergence). Large LR values on v or TB

values on u indicate shear. A smooth current field gives small kernel values, while a sudden jump or reversal between neighbouring cells gives a large value in at least one kernel.

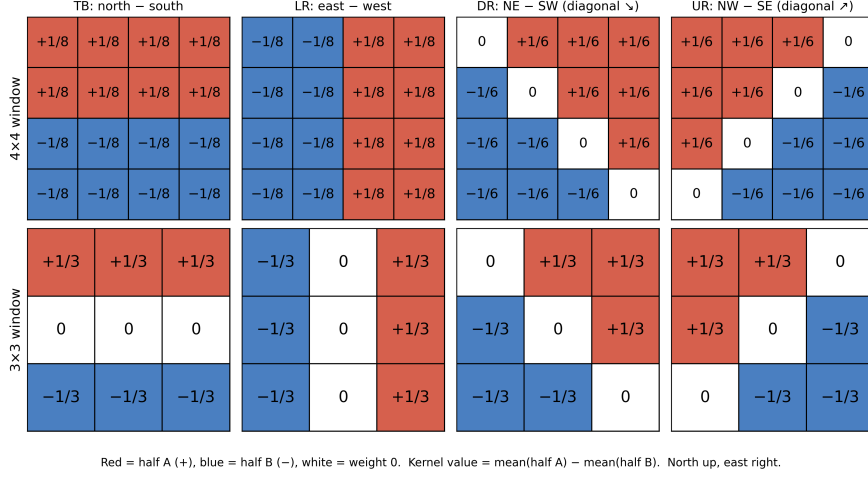


Figure 4: The four half-split kernels for the 4×4 (top) and 3×3 (bottom) windows. Each kernel returns the mean of half A (red, +) minus the mean of half B (blue, -); white cells have weight 0.

The *Isolation Forest* algorithm, an unsupervised anomaly-detection algorithm that needs no labelled examples, was used to find the windows whose combination of kernel values is unusual. The algorithm builds an ensemble of random binary trees: at each step it picks a random feature (here, a kernel value) and a random split value, and keeps dividing the data until each sample is isolated. Unusual samples lie far from the bulk of the data, so they are separated after only a few splits, whereas typical samples need many. The average number of splits over all trees is converted into an anomaly score, where a higher score means a more unusual window. The forest (200 trees) was trained on a random sample of up to 500,000 windows per product and then used to score every window in the 2021–2026 archive.

To make the selection concrete, the most anomalous events found by the sliding-window kernels and the *Isolation Forest* were reviewed visually. For each product, the five highest-scoring events for each window size (4×4 and 3×3) were selected, at least three days apart, giving 50 events in total. For every event, the current field was inspected together with the actual velocity values read from the netCDF files. Each event was classified as an error, a physical feature, or uncertain.

From this review, 31 reference cases were selected (Table 2). In 26 of them the field contains one of the target artefacts, although the existing QC algorithms rate all the cells involved as good.

The objective was for the new algorithm to flag these cases. The remaining cases were physical features or not artefacts, and the new algorithm must not flag them. These 31

cases were later used to tune and test the new flags.

Product	Divergence/convergence	Shear	Other anomalies
<i>Must be flagged</i>			
Total	2023-02-02 20:00, 2023-12-28 01:00, 2024-10-22 08:00, 2025-11-08 07:00	2024-02-02 08:00, 2023-12-31 04:30	2023-11-11 10:00*, 2023-06-21 07:00*
AURI	–	2022-03-29 11:30, 2026-02-25 01:00	2022-01-02 11:00
PIRA	2022-02-23 17:00, 2023-04-23 05:00, 2024-08-08 01:30	2025-07-09 20:30, 2022-03-23 07:30	2024-12-05 10:30
TRI1	2024-10-22 03:30, 2024-03-15 08:30	–	2024-06-11 17:00, 2024-03-03 19:30*
IZOL	2024-06-28 08:00, 2023-06-28 22:00, 2023-11-01 22:00, 2024-07-08 03:30	–	2024-04-08 23:00
<i>Must not be flagged</i>			
PIRA 2022-06-24 10:30; TRI1 2023-10-02 04:30; TRI1 2023-11-03 04:00 (front also seen by AURI and IZOL); Total 2024-10-01 17:00 (Bora jet); Total 2025-12-25, whole day (Bora storm).			

Table 2: Reference cases selected from the visual review (times in UTC).

* single-cell spike or reversal.

The following examples (Figures 5 - 9) illustrate the results of the aforementioned evaluation for both the 4×4 and 3×3 windows. One example is presented for each product, selected from one of the days listed in Table 2. Overall, the 3×3 windows performed better.

All figures follow the same layout.

For the radial products, the left-hand panel shows the QC-good radial velocity (RDVA) field and the corresponding radial velocity vectors. The pink box indicates the most anomalous 4×4 window, while the cyan box indicates the most anomalous 3×3 window, as identified by the Isolation Forest score. The right-hand panels show the values of the four kernels (rows: TB, LR, DR, and UR) applied to RDVA at each window position, first for the 4×4 windows and then for the 3×3 windows. Each kernel value is calculated as the difference between the mean radial velocities of the $+$ and $-$ halves (m/s). White indicates agreement between the two halves, whereas dark red and dark blue indicate large differences in radial velocity between neighbouring cells.

For the Total product, the left-hand panel shows the QC-good current velocity field. The pink and cyan boxes indicate the most anomalous 4×4 and 3×3 windows, respectively, as identified by the Isolation Forest score. The right-hand panels show the values of the four kernels (rows: TB, LR, DR, and UR) applied separately to the zonal (u) and meridional (v) velocity components at each window position, first for the 4×4 windows and then for the 3×3 windows. Each kernel value is calculated as the difference between the mean ve-

locities of the + and - halves (m/s). White indicates agreement between the two halves, while red and blue indicate that the + half has a higher or lower velocity component, respectively, than the - half. Specifically, red values for the TB kernel applied to v and the LR kernel applied to u indicate divergence, whereas blue values indicate convergence.

AURI — kernels over the whole map at 2022-03-29T11:30 UTC (4x4 top)

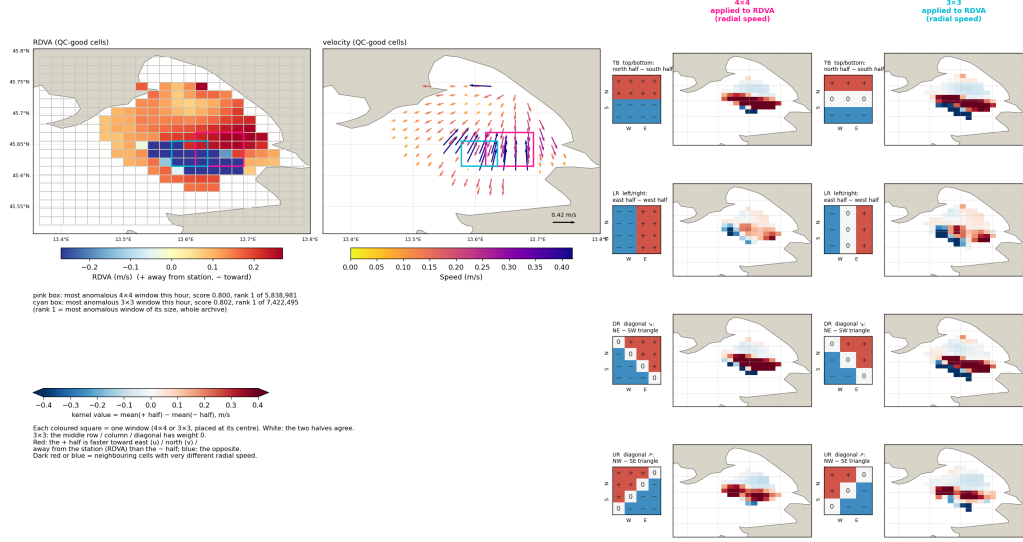


Figure 5: Kernel feature maps for the AURI radial product on 2022-03-29.

TRI1 — kernels over the whole map at 2024-10-22T03:30 UTC (3x3 top)

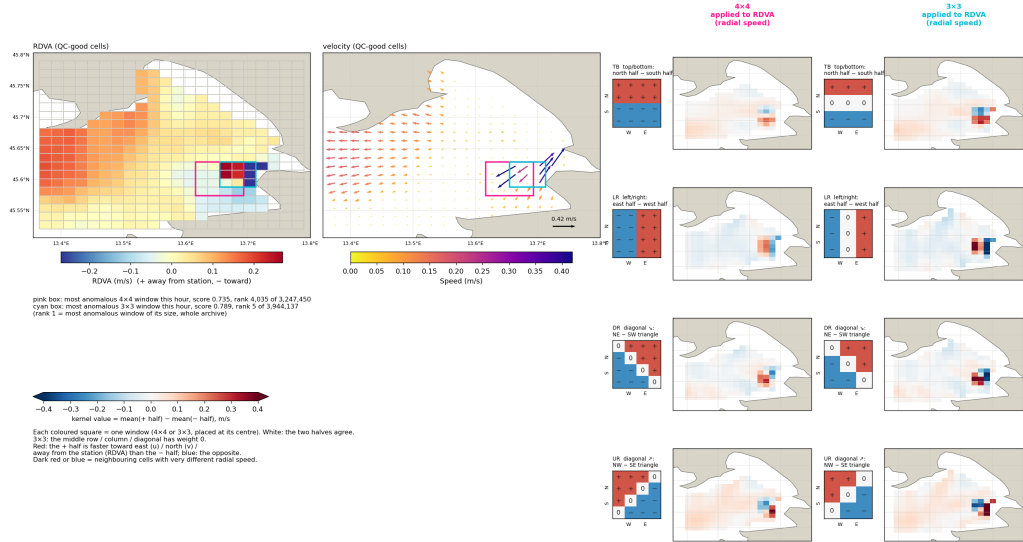


Figure 6: Kernel feature maps for the TRI1 radial product on 2024-10-22.

IZOL — kernels over the whole map at 2024-07-08T03:30 UTC (3x3 top)

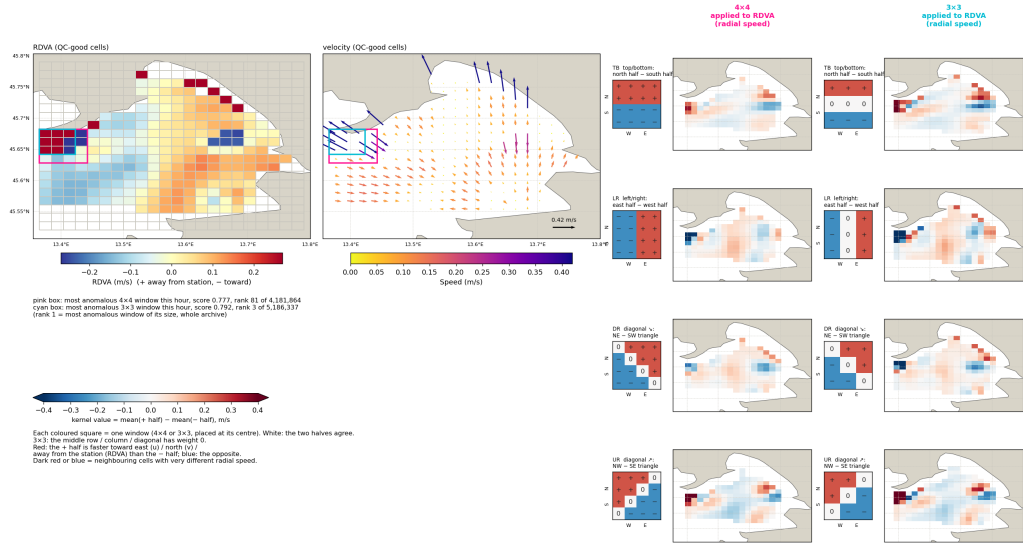


Figure 7: Kernel feature maps for the IZOL radial product on 2024-07-08.

PIRA — kernels over the whole map at 2024-08-08T01:30 UTC (3x3 top)

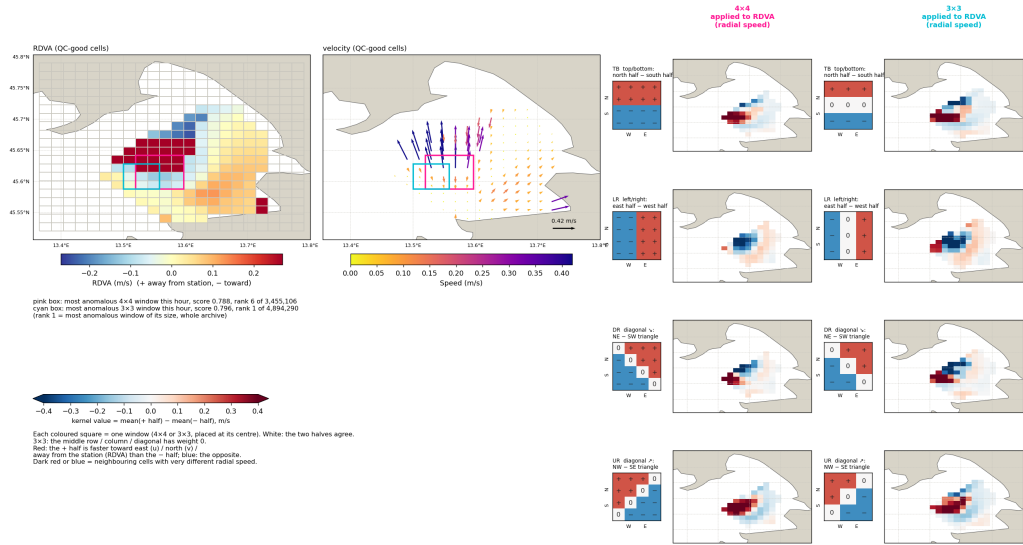


Figure 8: Kernel feature maps for the PIRA radial product on 2024-08-08.

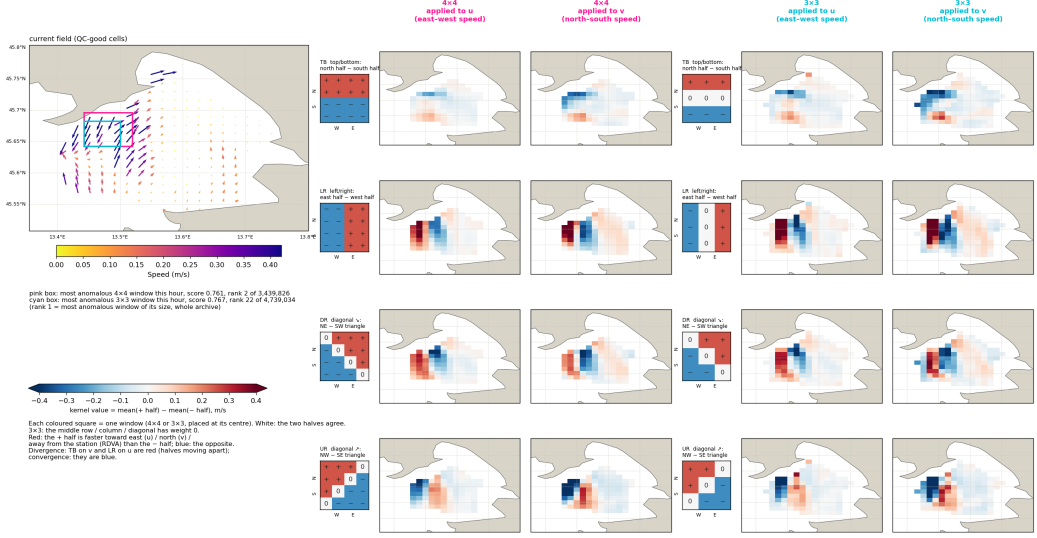


Figure 9: Kernel feature maps for the Total product on 2024-10-22.

Kernel-based QC flag

The anomaly score of the Isolation Forest identifies unusual windows, but an unusual window is not necessarily an error: a strong Bora jet next to calmer water is also unusual, yet physically real.

To turn the kernel results into a quality-control test, a flag that detects unusual windows, sorts them by type, and checks them against the other stations was built on top of them. The flag uses only cells marked as good by the existing EU HFR Node QC. It therefore complements the existing QC rather than replacing it.

Design choices

The design of the flag was guided by three preliminary tests on the reference cases selected (Table 2).

Firstly, two ways of selecting suspicious windows were compared: a physical threshold on the largest absolute kernel value in the window (criterion A) and a threshold on the Isolation Forest score (criterion B), both set at the same top fraction of the archive. For the radial products the two criteria selected the same windows in 87–89% of cases. For the total product they agreed in only 49% of cases, because criterion A selects windows with one very large jump, while criterion B selects windows that are moderately unusual in several kernels at once. At the top 0.5%, criterion A detected all reference cases and criterion B detected 23. A window is selected if it satisfies criterion A or criterion B.

Secondly, a reversal (neighbouring cells flowing in opposite directions, the target type) and an outburst (a region of strong flow next to weaker flow in the same direction) needed to be distinguished.

Comparing the means of the two halves of the window did not work for the radial

products, because a single opposite cell does not change the sign of a three-cell mean. Comparing every pair of neighbouring cells inside the window did work: a single pair of cells with opposite directions is enough to mark the window as a reversal. With this rule, the reference artefacts were classified as reversals, while strong but coherent flows, such as the Bora cases, were classified as outbursts.

Lastly, the third test examined whether a radial reversal can be confirmed by the other stations. Because all five products share the same grid and the same half-hourly time steps, the same location can be checked in the other radial products at the same time.

Flagging rule

The flag is computed in four steps.

In the first step, a complete 3×3 window (all nine cells good) is selected as suspicious if its largest absolute kernel value exceeds the threshold thr_A or its Isolation Forest score exceeds the threshold thr_B . Both thresholds are defined as the top 0.5% of the corresponding values over the whole 2021-2026 archive of each product (Table 5).

In the second step, each selected window is classified as a reversal or an outburst. A window is a reversal if at least one pair of neighbouring cells points in opposite directions, i.e. the angle between the two velocity vectors exceeds 90° (total product) or the two radial velocities have opposite signs (radial products). In addition, both cells must have a speed of at least $s_{\min}$, and their difference must be at least 0.3 m/s. All other selected windows are classified as outbursts. Outbursts are not treated as errors and remain flagged as good.

In the third step, each reversal in a radial product is checked against the other three radial stations within the window extended by one cell on each side (5×5 region) at the same time step. A station shows a jump if two neighbouring cells inside this region differ by at least 0.3 m/s. If at least two of the other stations show a jump, the reversal is considered a real feature and the cells remain good. If none of them shows a jump and at least one station has a smooth field there (at least five good cells), the reversal is contradicted by the independent observations and the cells are flagged as bad (flag 4 in Table 3, wp2). In all other cases, i.e. when only one other station shows a jump or no other station has data, the cells are flagged as probably bad (flag 3).

In the fourth step, reversals in the total product are always flagged as bad (flag 4). The total product is computed from the radial velocities, so the radial stations cannot confirm a total-product feature independently. Instead, the radial stations that show a jump at that location are recorded as the probable source of the reversal.

Finally, all nine cells of a flagged window receive the result of that window. When a cell belongs to several flagged windows, the most severe flag is kept.

Output variables

The flag produces three variables per grid cell and time step (Table 3).

`KREV_QC` is the QC itself and follows the same convention as the existing EU HFR Node QC flags.

`KOUT_FLAG` marks outbursts and confirmed real features. It is not a quality rejection; it

is intended for a possible future climatological analysis, where such strong events can be studied separately.

KREV_SRC is defined only for the total product and records which radial station or stations show a jump at the location of a total-product reversal.

Variable	Values	Meaning
KREV_QC	0, 1, 3, 4	0 = not tested (cell not inside any complete 3×3 window), 1 = good, 3 = probably bad, 4 = bad
KOUT_FLAG	0, 1	1 = outburst or reversal confirmed by at least two other stations (informational)
KREV_SRC	station list	total product only: radial station(s) showing a jump at a total-product reversal

Table 3: Output variables of the kernel-based QC flag.

Parameters

The parameters of the flag are listed in Table 4. They were first set from the preliminary tests and then tuned on the reference cases.

The resulting thresholds thr_A and thr_B for each product are given in Table 5.

For example, a thr_A of 0.40 m/s for the total product means that in the top 0.5% of all windows, the two halves of at least one kernel differ by at least 0.40 m/s.

Parameter	Value	Meaning
Top fraction	0.5%	thr_A and thr_B are set at the top 0.5% of each product’s archive
$s_{\min}$, total	0.15 m/s	Minimum speed of both cells in a reversal pair
$s_{\min}$, radials	0.05 m/s	Minimum absolute radial velocity of both cells in a reversal pair
Angle	90°	Total product: two vectors are opposite above this angle
Minimum difference	0.3 m/s	Minimum velocity difference within a reversal pair
Jump at other stations	0.3 m/s	Minimum difference between neighbouring cells at another station
Minimum cells	5	Number of good cells required at another station in the 5×5 region for it to count as smooth
Extension	1 cell	The 3×3 window is extended to a 5×5 region for the cross-station check

Table 4: Final parameters of the kernel-based QC flag.

Product	thr_A [m/s]	thr_B (score)
Total	0.40	0.675
AURI	0.22	0.720
PIRA	0.30	0.729
TRI1	0.23	0.710
IZOL	0.33	0.728

Table 5: Detection thresholds per product.

4 Validation of the new algorithms on representative sections of the 2021–2026 time series and adjustment of the parameters

For this analysis, the scripts *kernel_qc_flag.py*, *kernel_qc_report.py*, and *plot_qc_flagged_field.py* were developed.

Validation design

The new method was validated at three levels (Table 6).

First, it was checked on the whole 2021–2026 archive that the kernels and the Isolation Forest score measure what they are intended to measure.

Second, the flag was tuned and tested on the 31 reference cases (Table 2).

Third, the flag was applied to three complete months, chosen to represent different conditions, in order to assess its behaviour independently of the reference cases. The flag was computed for every day containing a reference case and for the three validation months.

Part	Data	Purpose
Kernels and anomaly score	Whole 2021–2026 archive, all five products	Check that the features and the score detect the intended anomalies
Reference cases	31 cases (26 must be flagged, 5 must not)	Tuning of the parameters and testing against labelled events
Validation months	December 2025 (winter, Bora); July 2026 (calm summer, all four stations); October 2024 (most anomalous month of the archive)	Assess behaviour under different conditions, independently of the reference cases

Table 6: Validation design.

Validation of the kernels and of the anomaly score

The most anomalous windows coincide with the anomalies targeted in this work.

In the radial products, the top 1% contain a pair of adjacent cells with opposite radial velocity 18 to 49 times more often than the average window. Strong but uniform flow is not considered anomalous.

The 3×3 windows performed better than the 4×4 windows in every check. They cover

30–40% more of the archive, because fewer cells have to be good at the same time. For this reason the QC flag uses 3×3 windows only.

In the total product, the anomalous windows are concentrated at the western edge of the coverage (13.37–13.42°E, 45.63–45.68°N), where about 3% of the windows are in the top 1%, while the open interior of the gulf stays at or below 1%. The same position is the strongest hot spot of IZOL station.

Parameter tuning

Each parameter of the flag was varied separately around its starting value, while the others were kept fixed (Table 7). For each setting, the following were counted: how many of the 26 cases that must be flagged received flag 3 or 4, how many of the 5 cases that must not be flagged remained good, and the share of tested cells flagged 3 or 4 in the three validation months. With the starting values, only the Christmas 2025 Bora day was handled incorrectly: two of its windows were classified as reversals with a minimum speed of 0.10 m/s. Raising the minimum speed for the total product to 0.15 m/s was the only single change that classified all 31 cases correctly, and it also reduced the share of flagged cells from 0.59% to 0.54%. This value was therefore adopted. The table also shows how sensitive the results are to each parameter.

Parameter	Values tested	Must be flagged	Must not	Share 3/4
Top fraction	0.25% / 0.5% / 1%	24 / 26 / 26	4 / 4 / 4	0.40 / 0.59 / 0.86%
Minimum difference	0.2 / 0.4 m/s	26 / 25	4 / 4	0.66 / 0.50%
$s_{\min}$, total	0.05 / 0.15 m/s	26 / 26	4 / 5	0.66 / 0.54%
$s_{\min}$, radials	0.03 / 0.08 m/s	26 / 24	4 / 4	0.68 / 0.49%
Angle	120° / 150°	26 / 25	4 / 4	0.52 / 0.46%
Jump at other stations	0.2 / 0.4 m/s	26 / 26	4 / 3	0.59 / 0.59%
Minimum cells	3 / 8	26 / 26	4 / 4	0.59 / 0.59%
Extension	0 / 2 cells	26 / 26	3 / 4	0.59 / 0.59%

Table 7: Sensitivity of the flag to each parameter.

With the final parameters, all 31 reference cases were classified correctly. All 26 cases that must be flagged are rated as good by the existing EU HFR Node QC, so each of them is an error that the existing QC misses and the new flag detects. The five cases that must not be flagged all remained good.

Cross-check

Table 8 shows the result of the cross-station check for the selected windows. In the radial products, most reversals were contradicted by the other stations (flag 4), and only a few were confirmed by at least two other stations (0–6 windows per station in three months). Real features seen by several stations are therefore rare compared with the reversals seen by only one station.

Product	Windows	Outburst	Contra- dicted (4)	One other station (3)	Not checkable (3)	Confirmed
AURI	510	452	39	11	8	0
PIRA	1,410	699	616	81	8	6
TRI1	2,957	2,562	303	78	11	3
IZOL	2,631	1,778	689	109	55	0
Total	3,486	2,731	755 (all flag 4)	—	—	—

Table 8: Cross-station check. “Not checkable” = no other station has data at that location.

For the total product, the flag records which radial stations show a jump at the location of each reversal. Of the 755 reversals in the validation months, 210 had a jump only in IZOL, 130 only in TRI1, 103 only in PIRA, 93 in two stations, and 2 only in AURI, while 208 had no jump in any radial and 9 had no radial data.

At the western edge of the coverage (longitude below 13.45°E), IZOL showed a jump in 181 of the 310 total-product reversals of the whole validation set (58%). This could support a hypothesis that the recurring artefact at the western edge of the total product originates in the far-range radial velocities of IZOL station.

Figure 10 shows an example of this hypothesis: a diagonal strip of reversals in the south-western part of the field, most of them traced to IZOL.

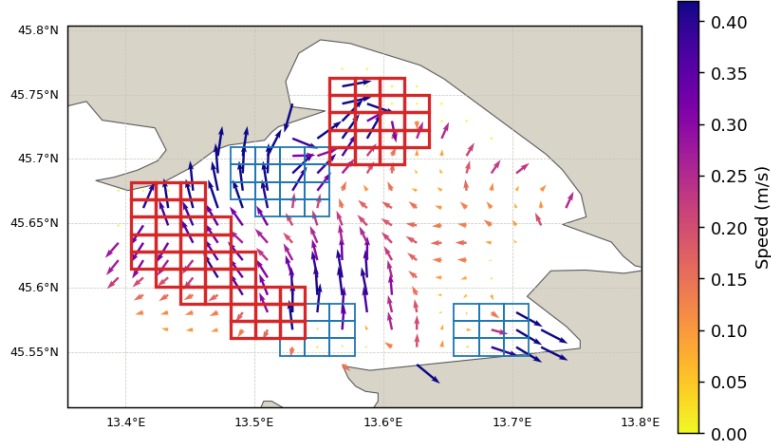


Figure 10: New QC flag in the total product, 2024-10-15. Red squares = flag 4, blue squares = outburst. The diagonal strip of reversals in the south-west was traced mainly to IZOL (8 windows), with one window traced to PIRA and two without a jump in any radial.

Figure 11 shows a radial example in which all possible results of the cross-station check occur in the same field.

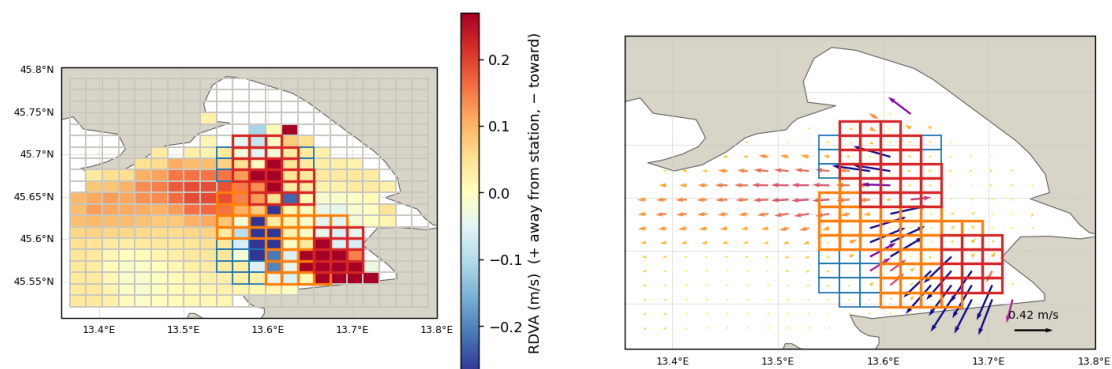


Figure 11: New QC flag for TRI1, 2024-10-26: radial velocity (left) and velocity vectors (right). Red squares = flag 4 (contradicted by the other stations), orange = flag 3 (seen by only one other station), blue = outburst or confirmed by at least two other stations. In this field 17 windows were classified as outbursts, 3 were confirmed as real, 15 were seen by one other station, and 7 were contradicted.

Comparison with the existing methods

Table 9 compares the new flag with the existing EU HFR Node QC. The existing QC evaluates single cells or whole scans (speed, variance, number of vectors, geometry, land), and it has no test for the physical consistency of neighbouring vectors. As a result, all 26 artefacts of the reference set pass as good data, while genuine Bora extremes are partly rejected by the speed threshold. The new flag detects all 26 artefacts, leaves the Bora cases and the real front as good, works on all five products, and identifies the radial station from which a total-product reversal most probably originates.

	Existing EU HFR Node QC	New kernel QC flag
Scope	single cells or whole scans: speed, variance, count, geometry, land	3×3 neighbourhood in all five products, plus the other stations at the same time
Unphysical divergence and shear	not detected (26 of 26 reference artefacts rated good)	26 of 26 reference artefacts detected; at most 1.25% of tested cells flagged 4
Bora extremes	partly rejected by CSPD.QC (example: 66.7% of the cells on 25 Dec 2025, 11:00)	not flagged
Real fronts	–	example: TRI1 front on 2023-11-03 confirmed by other stations
Origin of total-product errors	–	radial station(s) recorded in KREV_SRC

Table 9: Comparison of the new flag with the existing methods.

5 Conclusions

A kernel-based QC flag was developed that detects all 26 reference artefacts missed by the existing QC, leaves real Bora events and fronts as good, flags at most 1.25% of tested cells as bad, and traces total-product errors to their source radial station.

Limitations

The validation in the previous section showed the limits of the method:

- only cells inside a complete 3×3 window of good cells can be tested, so sparse areas and the edges of coverage gaps remain untested,
- a single cell that deviates from its neighbours without changing sign is diluted in the half means and is usually not detected, whereas a single cell with the opposite sign is detected,
- a radial station cannot see a velocity jump perpendicular to its beam, so a real front may appear smooth to one station; with two confirmations required, a real front covered by only two stations remains flagged as probably bad,
- reversals in the total product are never cleared, because the radial products from which it is computed cannot confirm it independently,
- the parameters were tuned on the same reference cases on which they were tested, so the result on the reference cases is a fit rather than an independent score,
- the validation covers only the 2021-2026 period available from wp1, and some validation months are affected by station outages.

Possible different kernel shapes

The four kernels used in this work all compare two halves of the window. This shape is well suited to fronts and to patches flowing against their surroundings, but the review and the validation showed two situations in which it is less effective. A single deviating cell contributes only one third of a half in a 3×3 window, so a spike of the same sign as its neighbours is diluted and usually missed.

A thin line of deviating cells that runs diagonally or along the centre of the window falls partly on both sides of the split and is weakened in the same way.

Two additional kernel shapes could address these cases: a *spike* kernel, which compares the centre cell with the mean of its eight neighbours, and *stripe* kernels, which compare a central row, column or diagonal with the cells on both sides of it. Both can be computed on the same 3×3 windows and added as further features to the Isolation Forest without changing the rest of the method.

Can unsupervised algorithms separate artefacts from other situations?

The k-means clustering on whole current fields separated the data into flow regimes, but

the artefacts averaged out in the cluster means, because they occur at a different place each time. Therefore the Isolation Forest algorithm was implemented. The Isolation Forest is an unsupervised method, which found the unusual windows: the most anomalous 1% contain adjacent cells with opposite radial velocities 18 to 49 times more often than average, and they include all of the reference artefacts. However, it cannot decide on its own whether an unusual window is an artefact or a real feature. Hence, unsupervised algorithms such as this are well suited to *finding* anomalies without labelled examples, but on their own they do not separate artefacts from physical situations. In the method proposed here, this separation is achieved by combining the unsupervised detection with a physically motivated rule: the classification into reversals and outbursts, and the consistency check with the other radar stations.